MATHS AA SL · Rates of change and accumulation

Worked mark scheme · 35 minutes · 28 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

1. Zooming in on a filling tank [7]

(a) **[2]**

Calculate the average rate for $h = 2$. Give units.

M1 · Use the change in depth divided by 2.

A1 · $(20.54 - 17.5)/2 = 1.52$ cm/s.

(b) **[3]**

Show that the average rate for any non-zero admissible h is $1.4 + 0.06h$. Hence find the instantaneous rate.

M1 · Expand $d(5 + h) - d(5)$.

A1 · Obtain $1.4h + 0.06h^2$ and divide by h.

A1 · The limiting rate is 1.4 cm/s.

(c) **[2]**

For positive h, find the largest interval length that gives an estimate within 0.015 cm/s of the instantaneous rate.

M1 · Require $0.06h \leq 0.015$.

A1 · $h \leq 0.25$ s.

2. Matching the speed at a join [7]

(a) **[4]**

Find a and b so that displacement and velocity are continuous at 2 seconds.

M1 · Match displacements at 2.

A1 · $a = 8$.

M1 · Match the derivatives $2t + 2$ and $b - 2(t - 2)$ at 2.

A1 · $b = 6$.

(b) **[3]**

With those values find the acceleration immediately before and after the join. Does continuous velocity guarantee continuous acceleration here?

A1 · Before the join the acceleration is 2 m/s^2.

A1 · After it the acceleration is −2 m/s^2.

R1 · The change in slope of velocity gives a jump in acceleration despite matching velocities.

3. When a tank is fullest [8]

(a) **[3]**

Find the amount in the tank at 5 minutes.

M1 · Area from 0 to 2 is $(2+6)\times 2/2 = 8$ litres.

M1 · From 2 to 5 it is $6\times 3/2 = 9$ litres.

A1 · The tank contains 37 litres.

(b) **[3]**

Find the amount at 8 minutes. Distinguish net change from total absolute flow.

M1 · Negative area is $4+2=6$ litres.

A1 · Final amount is 31 litres.

R1 · Net change is 11 litres, while 23 litres flowed in or out altogether.

(c) **[2]**

State when the tank is fullest and justify using the graph.

A1 · It is fullest at 5 min.

R1 · The rate changes sign from positive to negative there.

4. A calibration limit [6]

(a) **[3]**

Find the times for which both conditions hold.

M1 · Differentiate to obtain $12-6t$.

M1 · Solve $12-6t\ge 3$, which also implies an increasing output.

A1 · $0\le t\le 1.5$ s.

(b) **[3]**

A straight-line estimate from $t=1$ to $t=2$ has slope 3. Explain why the whole interval cannot be accepted.

M1 · The mean slope is $(42-39)/(2-1)=3$.

R1 · The instantaneous slope decreases over the interval.

A1 · For $1.5<t\le 2$ it falls below 3.

End of mark scheme · 28 marks